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Experimentno:04 Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

Programs:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
ann = MLPClassifier(
    hidden_layer_sizes=(5,),
    activation='relu',
    solver='adam',
    max_iter=1000,
    random_state=42
)
```

```
ann.fit(X_train, y_train)
```

```
y_pred = ann.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Actual Values:")
```

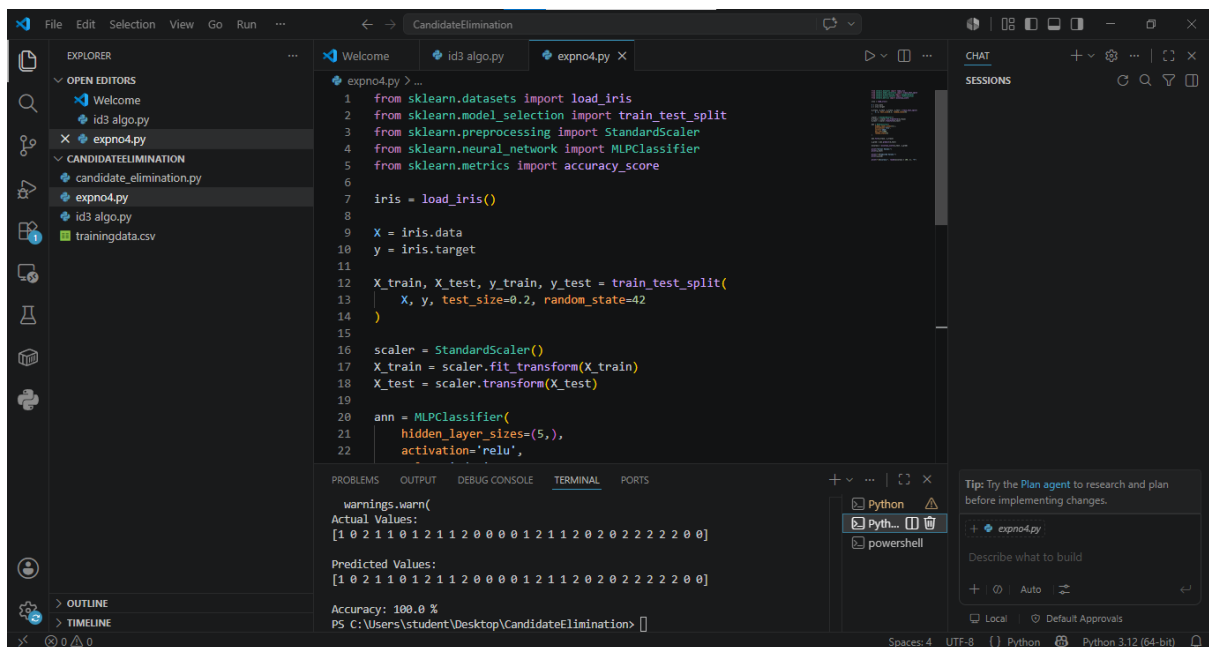
```
print(y_test)
```

```
print("\nPredicted Values:")
```

```
print(y_pred)
```

```
print("\nAccuracy:", round(accuracy * 100, 2), "%")
```

output:



```
1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.preprocessing import StandardScaler
4 from sklearn.neural_network import MLPClassifier
5 from sklearn.metrics import accuracy_score
6
7 iris = load_iris()
8
9 X = iris.data
10 y = iris.target
11
12 X_train, X_test, y_train, y_test = train_test_split(
13     X, y, test_size=0.2, random_state=42
14 )
15
16 scaler = StandardScaler()
17 X_train = scaler.fit_transform(X_train)
18 X_test = scaler.transform(X_test)
19
20 ann = MLPClassifier(
21     hidden_layer_sizes=(5,),
22     activation='relu',
```

warnings.warn(
Actual Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

Predicted Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

Accuracy: 100.0 %
PS C:\Users\student\Desktop\CandidateElimination>